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Research Article

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Enhanced recombinase polymerase amplification via UvsX engineering and reaction optimization for rapid detection of *Vibrio parahaemolyticus*

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Abstract

Recombinase polymerase amplification (RPA) is a powerful isothermal nucleic acid amplification technique, yet its efficiency is critically dependent on the catalytic efficiency of the recombinase UvsX, a key enzyme mediating homologous DNA pairing and strand exchange. To address this limitation, in this study, we developed a specific, sensitive, and robust RPA detection method by optimizing the UvsX enzyme through protein engineering and refining the RPA reaction system. By conducting comparative structural and functional analysis of UvsX orthologs from 13 Myoviridae phages, we identified critical determinants of recombinase activity within the Loop 2 domain of T4 UvsX. Furthermore, we systematically optimized the stoichiometric ratios of core enzymes and crowding agents to establish a robust RPA system. This system was subsequently integrated with lateral flow strips for point-of-need detection of highly lethal *Vibrio parahaemolyticus* in shrimp. Our results demonstrated that the engineered UvsXv1 variant exhibited significantly improved strand displacement activity, leading to enhanced RPA amplification efficiency and stability.

Impact Statement

The optimized UvsXv1–RPA–LF assay achieved sensitive detection of highly lethal *Vibrio parahaemolyticus* genomic DNA at concentrations as low as 1 pg μl^{-1} within 15 min, with exceptional specificity. Our work not only provides a field-deployable solution for foodborne pathogen surveillance but also establishes a generalizable framework for optimizing recombinase polymerase amplification technologies.

Keywords: recombinase polymerase amplification (RPA); recombinase UvsX; reaction optimization; *Vibrio parahaemolyticus*; rapid detection

Introduction

Nucleic acid amplification technology plays a pivotal role in biological science research, molecular diagnostics, forensic identification, and environmental monitoring (Kang et al. 2022). These techniques can be broadly categorized into two types: variable-temperature amplification and isothermal amplification. Variable-temperature nucleic acid amplification, primarily represented by PCR and its derivative technologies, requires precise instrumentation and specialized operational expertise, limiting its applicability in low-resource settings outside laboratory environments (Zhu et al. 2020). Isothermal amplification techniques, e.g. loop-mediated isothermal amplification, helicase-dependent amplification, and recombinase polymerase amplification (RPA), have emerged as promising alternatives, offering instrument-free operation and field-deployability (Li and Macdonald. 2015, Srivastava et al. 2023). Among these, RPA demonstrates unique advantages due to its minimal thermostatic requirements.

RPA mimics intracellular nucleic acid replication by utilizing UvsX, UvsY, gp32, and Bsu or Sau DNA polymerase, with T4 phage DNA replication serving as its prototype. This technique represents the most promising isothermal nucleic acid amplification method for potentially replacing PCR (Piepenburg et al. 2006, Lobato and O'Sullivan 2018, Li et al. 2019). The core component of RPA is the recombinase UvsX, a member of the RecA/Rad51 recombinase family that plays crucial roles in gene recombination, DNA damage repair, and replication (Yonesaki and Minagawa 1985). UvsX binds to single-stranded DNA (ssDNA) primers to form nucleoprotein filaments. Upon ATP activation, these filaments unwind double-stranded DNA (dsDNA) templates, search for homologous sites, and initiate strand displacement reactions, thereby facilitating nucleic acid amplification (Piepenburg et al. 2006). Consequently, UvsX activity represents a critical determinant of RPA performance.

In RPA, T4 UvsX is predominantly utilized (Piepenburg et al. 2006, Kojima et al. 2021). However, UvsX homologs are also present in other Myoviridae phages, including phages closely related to T4 and cyanophages (Millard et al. 2009). Piepenburg et al. demonstrated that UvsX derived from phages such as T6, RB69, Aeh1, and KVP40 can mediate RPA, although differences in their overall kinetic activities necessitate further modification and optimization (Piepenburg et al. 2022). To enhance UvsX performance, researchers have employed genetic engineering techniques. For instance, the T6 UvsX H66S variant exhibits higher activity and stability than

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the wild-type T6 UvsX (Piepenburg et al. 2022). Additionally, protein engineering advancements have led to the development of low-temperature-active UvsX variants by GenDx Biotech Co., Ltd (Suzhou, China), as well as the T4 UvsX F203Y&V244A variant designed by Jilin University, which extends the reaction temperature range to 20°C~45°C (Yu et al. 2023, Lü et al. 2024). These studies, primarily focused on T4 and its closely related phage-derived UvsX, provide valuable insights and novel strategies for UvsX optimization, further enhancing RPA's versatility and applicability in diverse settings. Previous RPA optimization efforts have primarily focused on reaction additives, including betaine, PEG 200, and nanoparticles, which improved specificity or efficiency to varying degrees (Luo et al. 2019, Tian et al. 2019, Tao et al. 2022, Kojima et al. 2021). However, such strategies often neglect systematic engineering of the core recombinase UvsX, limiting further enhancement of RPA performance.

Outbreaks of highly lethal *Vibrio* disease, characterized by empty digestive tracts and hepatopancreatic discoloration, have caused significant economic losses in China (Yang et al. 2022). Current PCR-based diagnostics require specialized laboratory equipment, which limits their use in resource-limited settings and delays on-site disease management (Liu et al. 2023, Jia et al. 2024). In this study, we implemented a comprehensive optimization strategy involving protein engineering of T4 UvsX based on phage-derived sequences and subsequent determination of its optimal reaction conditions to maximize RPA efficiency and specificity. This work not only introduces a novel approach for UvsX modification but also advances RPA optimization techniques. Moreover, we successfully developed an RPA-LF (lateral flow) visual detection platform and validated its application for detecting highly lethal *V. parahaemolyticus* (Vp-HLV). This technological advancement offers a powerful tool for monitoring and controlling outbreaks caused by these pathogenic *Vibrio* species.

Materials and methods

Bacterial strains

All strains were maintained at −80°C and cultured in a constant-temperature shaker. *Aeromonas salmonicida* subsp. *masoucida* (AS02) was grown in trypticase soy broth at 30°C, while highly lethal *V. parahaemolyticus* (Vp-HLV), *Vibrio* causing acute hepatopancreatic necrosis disease (AHPND; Yang et al. 2022), *V. parahaemolyticus* (RIMD2210633), and *Escherichia coli* BL21(DE3) were grown in lysogeny broth (LB) at 37°C.

Expression and purification of UvsX

The amino acid sequence of bacteriophage T4 recombinase UvsX (UniProt ID: P04529) was obtained from the UniProt database (https://www.uniprot.org/). A 6 × His-tag was incorporated at the N-terminus of the sequence, and the gene was synthesized and cloned into the pET-28a vector (Supplementary Sequence 1) by GenScript (Nanjing, China). The recombinant plasmid was then transformed into *E. coli* BL21(DE3) to generate the recombinant strain, followed by verification through sequencing. Cells were grown in LB medium containing 50 μg ml^{-1} kanamycin to OD_{600} 0.6~0.8 before induction with 1 mmol l^{-1} isopropyl β-D-1-thiogalactopyranoside at 20°C for 18 h.

The cells were harvested by centrifugation at 8000 × *g* for 5 min at 4°C, (2~4 g wet weight l^{-1}). The harvested cells were then resuspended in buffer A (20 mmol l^{-1} Tris–HCl, 500 mmol l^{-1} NaCl, pH 7.8) at 1:10~1:15 (w/v) and lysed by sonication (60% amplitude, 4 s on/8 s off, 10 min; two cycles). After centrifugation at 15 000 × *g* for 30 min at 4°C, the supernatant was filtered through a 0.45 μm membrane and applied to the column packed with Chelating Bestarose FF (Bestchrom, Shanghai, China) equilibrated with buffer A. The column was washed with 50 mmol l^{-1} imidazole, and UvsX was eluted using either a stepwise gradient (100, 200, 300, 400, and 500 mmol l^{-1} imidazole) or a linear gradient (100~400 mmol l^{-1} imidazole). Eluted fractions were analyzed by SDS-PAGE. Target fractions were dialyzed overnight at 4°C into storage buffer (50 mmol l^{-1} Tris–HCl, 50 mmol l^{-1} KCl, 500 mmol l^{-1} NaCl, 1 mmol l^{-1} DTT, 0.1 mmol l^{-1} EDTA, and 20% glycerol, pH 7.5) and stored at −80°C.

Preparation of DNA template

In this study, bacterial genomic DNA served as the standard template for RPA. Briefly, 2 ml of bacterial culture at the logarithmic growth phase was collected, and genomic DNA was extracted using the SteadyPure Bacterial Genomic DNA Extraction Kit (AG Biotech Co., Ltd, Hunan, China) following the manufacturer's protocol for Gram-negative bacteria. The extracted DNA was quantified using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA) and stored at −20°C.

Design and screening of primers

RPA primers were designed according to the TwistDX guidelines. The primers used in this study are listed in Supplementary Table 1. For *A. salmonicida*, the RPA primers were designed as previously reported by Zhang et al. (2024). For Vp-HLV, the RPA primers were designed to target the key virulence gene VTcA (Yang et al. 2023). To identify the optimal primer set with high specificity and sensitivity, gradient concentrations of Vp-HLV genomic DNA were used as templates for PCR screening.

Commercial UvsX-RPA

Commercial UvsX-RPA reaction system comprised the following components: 120 ng μl^{-1} UvsX (GenDx Biotech Co., Ltd), 30 ng μl^{-1} UvsY (HaiGene Biotech Co., Ltd, Harbin, China), 900 ng μl^{-1} gp32 (HaiGene Biotech Co., Ltd), 90 ng μl^{-1} Bsu DNA polymerase (GenDx Biotech Co., Ltd), 300 nmol l^{-1} primers, 200 μmol l^{-1} dNTPs, 5% PEG 35000, 2 mmol l^{-1} DTT, 20 mmol l^{-1} phosphocreatine, 3 mmol l^{-1} ATP, 50 mmol l^{-1} Tris–HCl (pH 7.9), 100 mmol l^{-1} potassium acetate, 100 ng μl^{-1} creatine kinase (Sigma-Aldrich), and 14 mmol l^{-1} magnesium acetate. The reaction mixture was supplemented with 1 μl of template DNA, thoroughly mixed, and incubated at 37°C for 1 h. The amplification products were stained with ethidium bromide and separated on a 2% agarose gel. The intensity of the target bands was quantified using ImageJ 1.54f.

Structure–function analysis of UvsX

The crystal structure of $UvsX_{(30–358)}$ was analyzed to provide insights into its structural features (Gajewski et al. 2011). However, no comprehensive or detailed structure–function studies have been reported. To identify potential functional domains in T4 UvsX, we referred to *E. coli* RecA (UniProt ID: P0A7G6), the most extensively characterized recombinase in terms of structure and function. The structural information of RecA (PDB ID: 2REB) was obtained from the Protein Data Bank (https://www.rcsb.org/). This analysis was performed through amino acid sequence alignment and advanced structural prediction. The full-length sequences of *E. coli* RecA and T4 UvsX were retrieved from UniProt database. Amino acid sequence alignment was performed using ClustalW in MEGA11 and visualized with ESPript 3.0. Based on key functional sites in RecA, we inferred the putative functional domains in T4 UvsX.

Activity assay of UvsX and its variants

Commercial UvsX was replaced with in-house purified UvsX and its variants in RPA to evaluate their activity. To further investigate the functional mechanism, the strand displacement activity of UvsX was characterized using the following two approaches. First, incubation experiments were conducted: the reactions containing dsDNA (the sequence is shown in Supplementary Sequence 2), 100 mmol l^{-1} potassium acetate, 1 mmol l^{-1} DTT, 10 mmol l^{-1} magnesium acetate, and 20 mmol l^{-1} Tris–HAc buffer were denatured (95°C, 3 min), and snap-cooled (ice, 10 min). UvsXv1 or an equal volume of deionized water was added, followed by incubation (37°C, 10 min). Initial reaction products (10 μl) were analyzed by 1% agarose gel electrophoresis. The remaining mixture was supplemented with ATP (3 mmol l^{-1} final concentration) and further incubated (room temperature, 30 min) prior to electrophoretic analysis. Second, a three-strand displacement assay was performed according to the method described by Namsaraev et al. (1998), and the results were compared with those of T4 UvsX.

Optimization of RPA reaction conditions

Critical enzyme ratio optimization. UvsXv1 was fixed at 120 ng μl^{-1}, and UvsY was added at concentrations ranging from 10 to 50 ng μl^{-1} for RPA. The amplified products were analyzed by agarose gel electrophoresis, and the optimal concentration ratio of UvsXv1 to UvsY was determined based on three replicate experiments. Based on the recommended UvsX concentration range (100∼200 ng μl^{-1}), the concentrations of UvsXv1 and UvsY were further refined. Subsequently, different concentrations of gp32 were tested in RPA to determine the optimal gp32 concentration.

Optimal crowding agents selection. RPA was performed using PEG 4000, PEG 6000, and PEG 35000 as crowding agents under conditions where the optimal concentrations of critical enzymes were fixed. The PEG concentration was varied from 0% to 10% (w/v), and the optimal type and concentration of the crowding agent were determined experimentally.

Sensitivity and stability analysis of UvsXv1-RPA

A systematic evaluation of UvsXv1-RPA sensitivity and stability was performed using commercial UvsX-RPA as the benchmark control. Sensitivity was determined via gradient dilutions of *A. salmonicida* genomic DNA (1∼10^6 fg μl^{-1}), while inter-batch stability was assessed using 1 ng μl^{-1} template. All reactions were conducted at 37°C for 1 h under standardized conditions. Amplification products were cryopreserved (−20°C) and finally analyzed by agarose gel electrophoresis together, with band intensities quantified using ImageJ 1.54f.

Statistical analysis

All experiments were performed in triplicate (n = 3). Agarose gel results were analyzed semi-quantitatively using ImageJ 1.54f. Contrast was optimized via ImageJ's Brightness/Contrast tool to ensure clear bands, no artifacts, and linear intensities. Images were in 8-bit TIFF format with background subtraction (50-pixel rolling ball). Data are shown as mean ± SD. Statistical significance ($P < .05$) was determined by Student's *t*-test or one-way ANOVA in GraphPad Prism 9.0.

Optimization of reaction temperature and time for UvsXv1–RPA–LF

To establish the optimal detection conditions, the RPA reactions were conducted at temperatures ranging from 30°C to 45°C, specifically 30°C, 35°C, 37,°C 39,°C 42°C, and 45°C, and incubated for 5, 10, 15, 20, 25, and 30 min. Following the completion of the reaction, the RPA amplification products were diluted 10-fold with PBS and subsequently analyzed using nucleic acid detection strips (Amp-Future Biotechnology Co., Ltd, Changzhou, China).

Specificity and sensitivity analysis of UvsXv1–RPA–LF

In this study, the UvsXv1–RPA–LF technique was employed to detect Vp-HLV. To evaluate the specificity of UvsXv1–RPA–LF, genomic DNA from *Vibrio* causing AHPND and *V. parahaemolyticus*, and plasmids containing the *Enterocytozoon hepatopenaei* specific gene encoding spore wall protein 2 were used as templates for RPA reactions. To validate detection specificity, we performed a BLASTn analysis of the target sequence against the NCBI nucleotide collection (nr/nt) database. Simultaneously, the sensitivity of UvsXv1–RPA–LF was analyzed by performing reactions with gradient dilutions of genomic DNA from Vp-HLV (1, 10, 10^2, 10^3, 10^4, 10^5, and 10^6 fg μl^{-1}).

Results and discussion

Sequence analysis and mutants design of UvsX

To identify critical functional sites of UvsX, we performed amino acid sequence alignment analysis using the well-characterized *E. coli* RecA protein as a reference (Fig. 1a). Although sequence identity between the two proteins was low (21.6%), they exhibited significant similarity at key functional sites. Based on previous studies of *E. coli* RecA (Del et al. 2019, 2021), we propose the following functional mechanism for UvsX in RPA: First, primers bind to ssDNA-binding sites (D156-P214) to form nucleoprotein filaments, with Loop 1 (D156-V163) and Loop 2 (T196-M208) being particularly important for DNA binding. Subsequently, ATP binds to the ATP-binding region (G60-S67), activating the nucleoprotein filaments. The activated UvsX then unwinds the dsDNA template and performs homology search, during which dsDNA interacts sequentially with the N-terminal domain (M1-E28) and the C-terminal domain (P268-E391) before transferring to the dsDNA-binding site (I224-N243). If ssDNA shows ho-

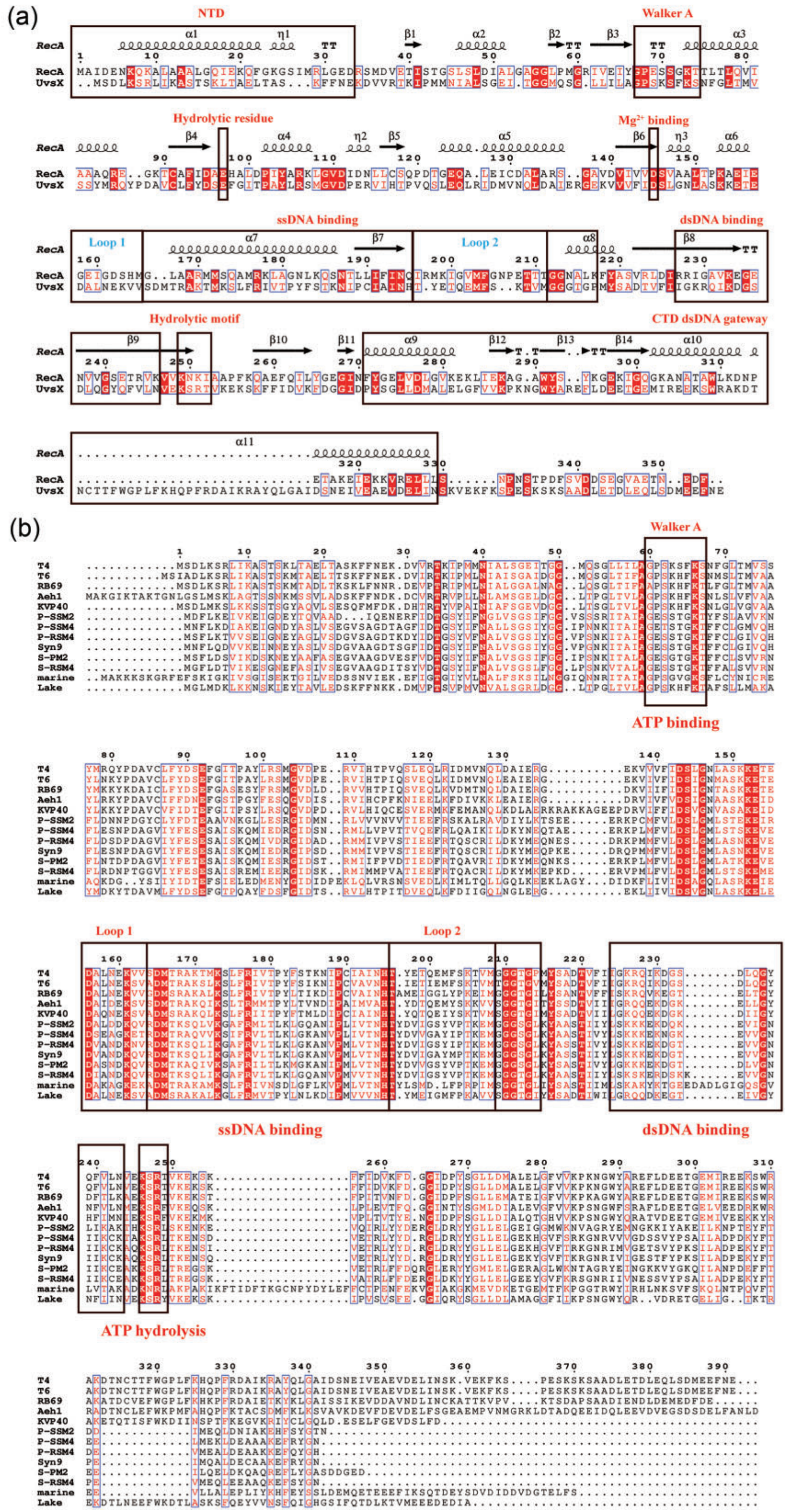

Figure 1. Sequence analysis of UvsX. Red background indicates identical amino acid residues, and red letters indicate similar amino acid residues. (a) Sequence alignment of UvsX and RecA. The secondary structure of RecA is annotated above the sequence alignment. (b) Multiple sequence alignment of UvsX proteins from 13 Myoviridae phages. Marine: uncultured marine phage. Lake: Lake Baikal phage Baikal-20–5m-C28.

mologous to dsDNA, strand exchange occurs accompanied by ATP hydrolysis; otherwise, dsDNA is released. Thus, UvsX's DNA affinity and ATP hydrolysis activity are key regulators for RPA performance.

We analyzed sequences from 13 representative Myoviridae phage-derived UvsX proteins to explore novel modification strategies, including T4 phage and its closely relatives as well as cyanophages (Fig. 1b). The results revealed that

Table 1. UvsX and its variants.

Name	The source of Loop 2 amino acid sequence	The specificity of species
UvsX	Enterobacteria phage T4	Standard
UvsXv1	*Synechococcus* phage S-RSM4	Prolonged exposure to high-intensity light
UvsXv2	Uncultured marine phage (GenBank: CAG7581408.1)	Failure to culture under laboratory conditions
UvsXv3	Lake Baikal phage Baikal-20–5m-C28	Adaptation to extreme nutrient deprivation

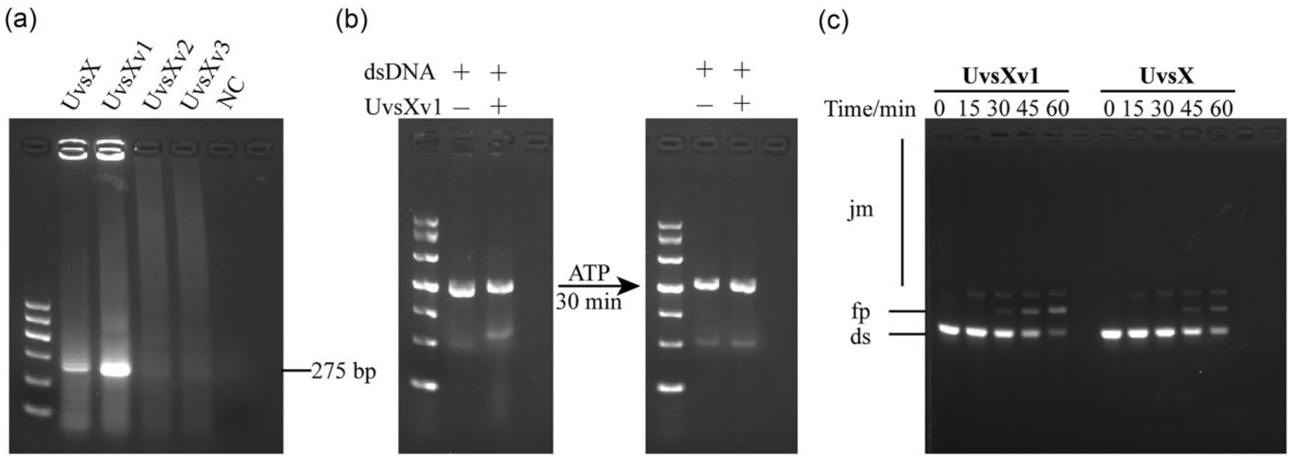

Figure 2. Functional activity analysis of UvsX and its variants. (a) UvsX and its variants in RPA. NC: Negative control. M: marker. (b) Characterization of strand displacement activity by UvsXv1. The negative control without recombinase showed no detectable band shift, while reaction containing UvsXv1 exhibited significant migration retardation (left, Lane 2). Normal electrophoretic mobility was restored upon ATP addition (right, Lane 2), confirming ATP-dependent strand displacement activity. (c) Comparative analysis of strand displacement activity between UvsX and UvsXv1. Strand displacement assays were performed using dsDNA of M13mp19 phage as substrate in standard reaction buffer (1 mmol l^{-1} ATP and its regeneration system, 10 μmol l^{-1} ssDNA of M13mp19 phage, 6 μmol l^{-1} UvsXv1 or UvsX, 0.9 μmol l^{-1} gp32, 50 mmol l^{-1} Tris–HCl, 10 mmol l^{-1} magnesium acetate, and 2 mmol l^{-1} DTT). jm: joint molecules (intermediate products); fp: final products (nicked heteroduplex DNA molecule); ds: dsDNA of M13mp19 phage (starting reactant).

sequences associated with ATP binding and hydrolysis were highly conserved among these UvsX proteins, while the Loop 2 region exhibited substantial variability, without significant positive charge accumulation. However, cyanophage-derived UvsX proteins displayed highly consistent sequence features in the Loop 2 region. Notably, cyanophage-derived UvsX proteins, which evolved under constant high-light exposure, may possess enhanced activity to support intensive DNA repair processes. Based on these findings, we selected the T4 UvsX Loop 2 region for modification, replacing it with corresponding sequences from three specific phage-derived UvsX proteins (Table 1).

Expression and activity analysis of UvsX mutants

Through protein purification, we successfully obtained wild-type T4 UvsX and three Loop 2-engineered mutants (Supplementary Fig. 1). Approximately 24 mg of the target protein was obtained per liter of culture medium. When tested in RPA, both wild-type T4 UvsX and UvsXv1 effectively mediated amplification, with UvsXv1 showing nearly 2-fold higher activity than wild-type T4 UvsX (Fig. 2a). In contrast, UvsXv2 and UvsXv3 failed to support efficient RPA, indicating that not all Loop 2 substitutions enhance UvsX activity.

To further explore structure–function relationships and improve DNA-binding affinity and ATP hydrolysis activity, additional UvsX variants were engineered. These included UvsX-R248K (modifying a key ATP hydrolysis residue), UvsX-Δ(N315-I333) (removing a C-terminal region absent in RecA), and C-terminal truncation mutants (UvsX-ΔC10, ΔC17, ΔC21, and ΔC35). Most mutants were successfully purified, but UvsX-Δ(N315-I333) showed reduced solubility (Supplementary Fig. 1). Subsequent RPA assays revealed that none of the soluble mutants maintained detectable or stable activity (Supplementary Fig. 2), indicating that RecA engineering strategies cannot be directly applied to UvsX. The successful enhancement of UvsXv1 suggests that future optimization efforts would benefit from high-resolution structural guidance.

Our studies further corroborate previous reports that Loop 2 substitutions enhance UvsX reaction performance (Piepenburg et al. 2022). To investigate the functional mechanism of UvsXv1, we characterized its strand displacement activity (Fig. 2b). Compared to the recombinase-free control, incubation with UvsXv1 resulted in significantly retarded bands, suggesting nonspecific, ATP-independent ssDNA binding. Subsequent ATP addition caused no further band shift, indicating UvsXv1 dissociated from ssDNA. DNA strand displacement assays confirmed that UvsXv1 initiated strand displacement more rapidly than wild-type T4 UvsX, yielding detectable products 33% faster (Fig. 2c). These results support a model wherein the Loop 2 modifications may enhance both DNA-binding affinity and strand invasion kinetics during RPA, although further validation is necessary.

While functional assays (RPA efficiency and strand displacement measurements) provide evidence of improved performance, they cannot fully elucidate the precise biophysi-

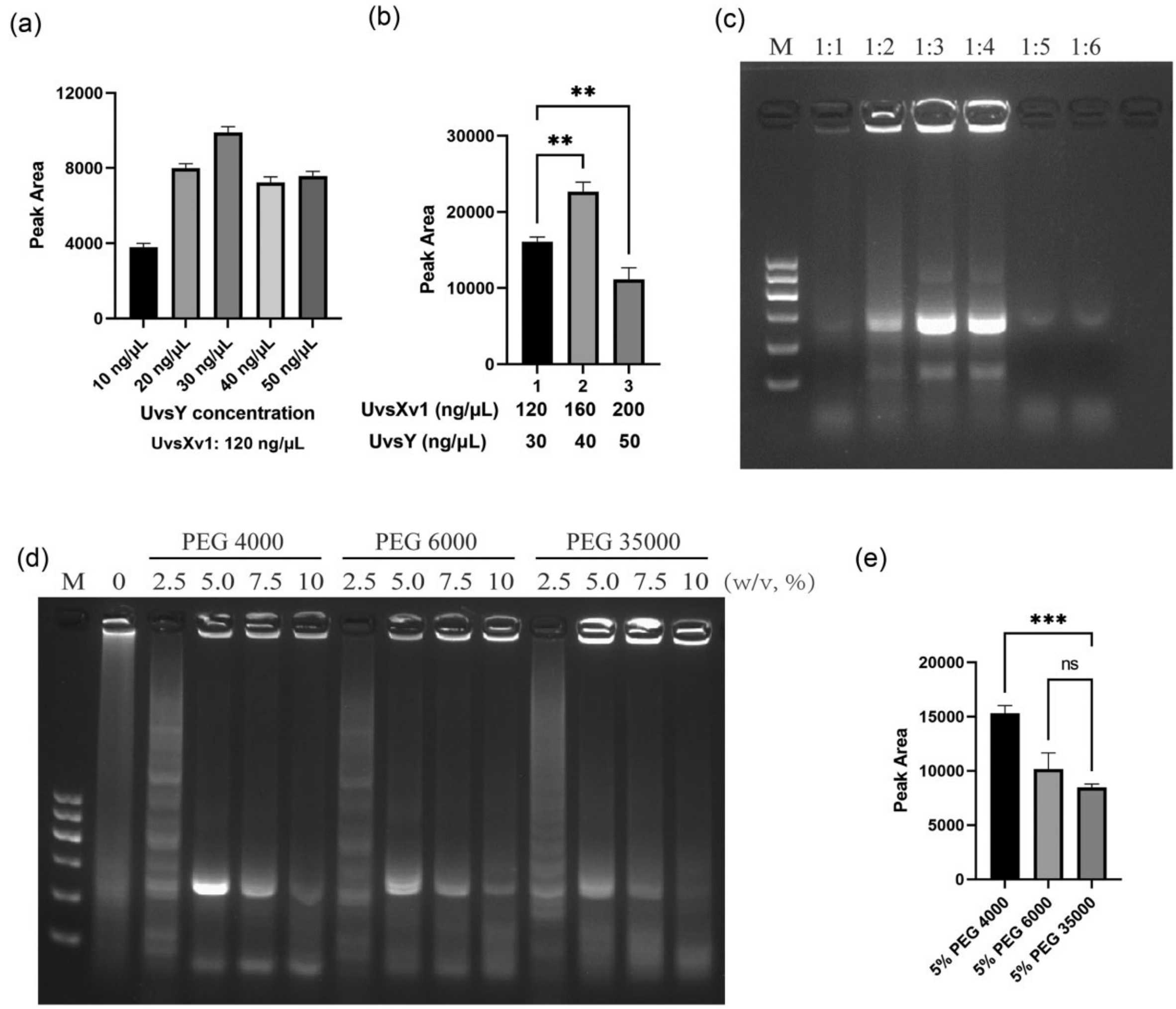

Figure 3. Optimization of reaction conditions for UvsXv1-RPA. (a) Optimization of UvsXv1:UvsY. Data represent mean ± SD ($n = 3$), and one-way ANOVA was employed for statistical analysis. (b) RPA reaction efficiency with UvsXv1:UvsY = 4:1. Data represent mean ± SD ($n = 3$). Statistical significance was determined using one-way ANOVA ($**P < 0.01$). (c) Optimization of UvsXv1:gp32 ratio. The concentration ratios indicated above each lane represent UvsXv1 to gp32. M: marker. (d) Optimization of crowding agents. (e) Quantitative analysis of crowding agents optimization. Data represent mean ± SD ($n = 3$), and statistical significance was determined using one-way ANOVA ($***P < 0.001$, ns: $P \geq 0.05$). The raw data are shown in Supplementary Fig. 3.

cal mechanisms underlying these enhancements. Future work should focus on elucidating the detailed biochemical mechanisms underlying the enhanced activity of UvsXv1, including quantitative ATPase kinetics, thermodynamic profiling of DNA binding, and filament stability, potentially through techniques such as cryo-EM, single-molecule FRET, and comprehensive biochemical assays. Implementation of qPCR-based absolute quantification will further replace current semi-quantitative ImageJ analysis, enabling standardized performance benchmarking.

Optimization of reaction conditions for UvsXv1-RPA

To explore the optimal application conditions of UvsXv1 in RPA and further enhance RPA performance, the reaction conditions were systematically optimized. First, we determined the optimal UvsXv1:UvsY concentration ratio as 4:1 (Fig. 3a). Further testing within recommended concentration ranges established optimal UvsXv1 and UvsY concentrations as 160 and 40 ng μl^{-1}, respectively (Fig. 3b). Finally, gp32 optimization showed good RPA efficiency at UvsXv1:gp32 ratios of 1:3 or 1:4 (Fig. 3c).

We next optimized crowding agents in the reaction system (Fig. 3d and e). The RPA reaction failed to achieve efficient amplification in the absence of crowding agents, confirming their essential role in the amplification process. Three polyethylene glycol (PEG) variants with different molecular weights were evaluated, all of which mediated RPA at an optimal concentration of 5% (w/v). Among these, PEG 4000-mediated RPA amplification yielded the highest efficiency and specificity. These results established the optimal UvsXv1-RPA conditions as 160 ng μl^{-1} UvsXv1, 40 ng μl^{-1} UvsY, 480 ng μl^{-1} gp32, and 5% (w/v) PEG 4000.

Performance comparison of UvsXv1-RPA

To comprehensively evaluate UvsXv1-RPA performance, we performed comparative analysis using *A. salmonicida* genomic DNA as a template against commercial UvsX-RPA. Both assays demonstrated equivalent sensitivity, detecting as little as 10 pg μl^{-1} of *A. salmonicida* genomic DNA (Fig. 4a).

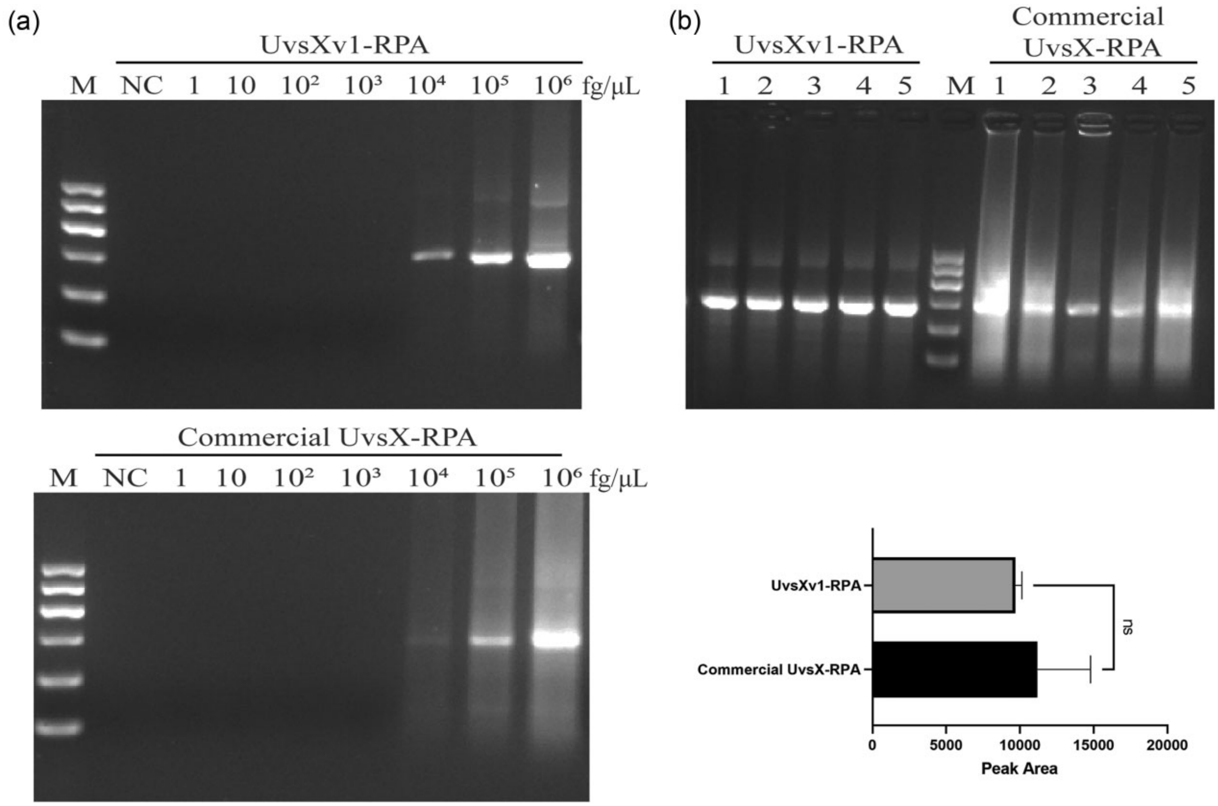

Figure 4. Comparative analysis of sensitivity and stability between UvsXv1-RPA and commercial UvsX-RPA. (a) Sensitivity analysis. RPA amplification of *A. salmonicida* genomic DNA with increasing template concentrations. NC: negative control. (b) Stability analysis. Lanes 1–5: the results of RPA from different batches. ns: $P \geq 0.05$. M: marker.

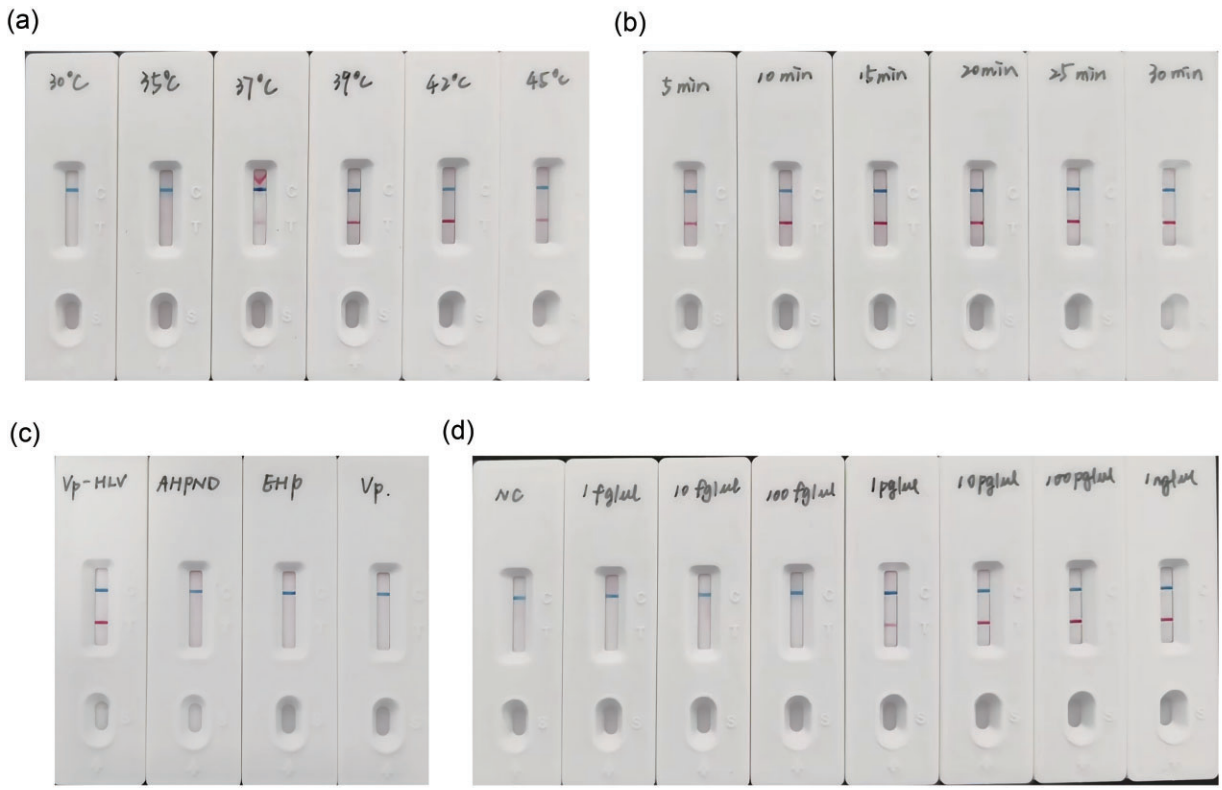

Figure 5. Rapid detection of Vp-HLV using UvsXv1–RPA–LF. (a) Optimization of reaction temperature. (b) Optimization of reaction time. (c) Specificity of UvsXv1–RPA–LF detection for Vp-HLV genomic DNA. (d) Sensitivity analysis of UvsXv1–RPA–LF detection for Vp-HLV genomic DNA. NC: negative control.

Notably, UvsXv1-RPA showed significantly reduced non-specific amplification compared to commercial UvsX-RPA. Replicate experiments confirmed that UvsXv1-RPA exhibited more consistent performance while maintaining comparable amplification efficiency to commercial UvsX-RPA (Fig. 4b). The performance improvements of UvsXv1-RPA hold significant potential for advancing RPA technology.

Rapid detection of Vp-HLV using UvsXv1–RPA–LF

For rapid field detection of Vp-HLV, we first screened and identified the HLV-F1 and biotin-labeled HLV-R1 as the optimal primer pair (Supplementary Fig. 4). Subsequent optimization of reaction parameters showed clear amplification bands between 37°C and 45°C, with optimal results at 42°C (Fig. 5a). Then, we tested the optimal reaction time for UvsXv1–RPA–LF at 42°C (Fig. 5b). When the RPA reaction was conducted for 5 min, a distinct band was observed, highlighting the high efficiency of UvsXv1–RPA–LF. Within the reaction time of 5∼15 min, the band intensity gradually increased over time. Based on these results, we determined that the optimal reaction conditions for UvsXv1–RPA–LF were 42°C and 15 min.

Finally, the specificity and sensitivity of UvsXv1–RPA–LF were evaluated. RPA reactions were performed at 42°C for 15 min. Specificity analysis demonstrated that a significant band was observed only when the Vp-HLV genome DNA served as the template, with no cross-reactivity observed, even with symptomatically similar AHPND *Vibrio*, confirming both high specificity and sensitivity for field applications (Fig. 5c). Despite the limited scope of specificity testing, the BLASTn analysis of the target sequence revealed its uniqueness to Vp-HLV (Supplementary Excel). These findings indicated that UvsXv1–RPA–LF was highly specific for the detection of Vp-HLV and did not cross-react with other pathogens. Furthermore, UvsXv1–RPA–LF successfully detected Vp-HLV genomes at concentrations as low as 1 pg μl^{-1} (equivalent to 177 copies μl^{-1}, Fig. 5d). UvsXv1–RPA–LF exhibited a 10-fold enhancement in detection sensitivity compared to UvsX–RPA–LF, confirming its superior analytical performance (Supplementary Fig. 5).

UvsXv1–RPA–LF has been successfully implemented for rapid detection of Vp-HLV, demonstrating significant potential for field-deployable diagnostics. However, the current study did not validate performance with clinical samples, and the long-term stability and reproducibility require further rigorous assessment. Future studies will validate the platform's robustness and expand its diagnostic utility through the incorporation of pathogen-specific probes for detection of diverse targets.

Conclusions

In summary, UvsX engineering and reaction optimization represent significant advances in RPA technology. The enhanced activity of UvsXv1 enables the development of sensitive, specific, and stable rapid detection methods, substantially expanding the potential application of RPA-based diagnostics.

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Author contributions

Lin Wang (Data curation [equal], Methodology [equal], Visualization [equal], Writing – original draft [equal]), Yiming Li (Data curation [equal], Visualization [equal]), Pengbo Wang (Supervision [equal], Validation [equal]), Yibei Zhang (Conceptualization [equal], Investigation [equal], Writing – review & editing [equal]), and Qin Liu (Conceptualization [equal], Investigation [equal], Writing – review & editing [equal])

Supplementary data

Supplementary data is available at *LAMBIO Journal* online.

Conflict of interest: The authors declare that they have no conflict of interest.

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Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

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